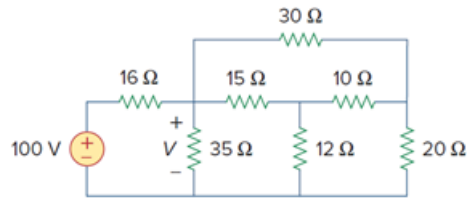


Problem

Determine V in the circuit of Fig. 2.120.

Figure 2.120:



Step-by-step solution

Step 1 of 6

Consider the following resistive circuit with delta abc marked as shown in Figure 1.

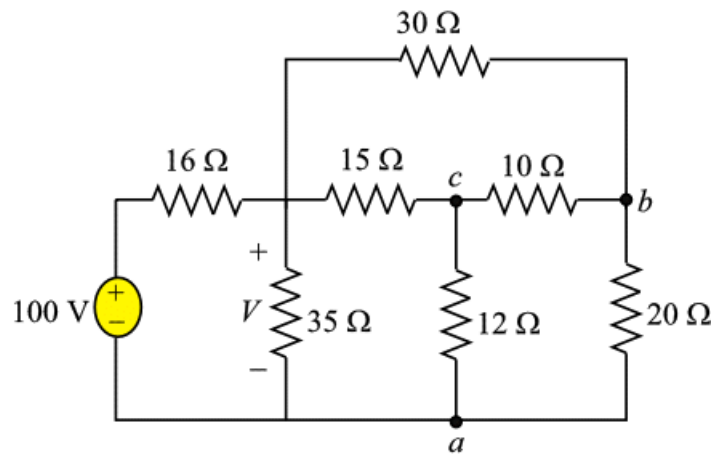


Figure 1

Step 2 of 6

Convert the delta abc to star network.

Redraw the circuit in Figure 1 by replacing the delta abc with star network.

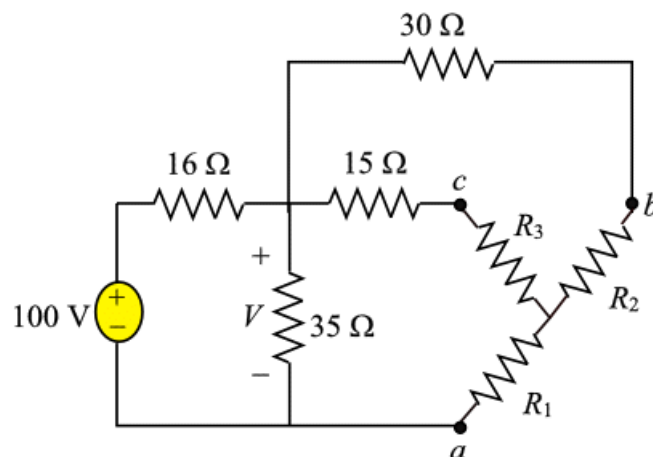


Figure 2

Step 3 of 6

Determine the equivalent Y resistances marked in the circuit shown in Figure 2.

$$\begin{aligned} R_1 &= \frac{(12)(20)}{12 + 20 + 10} \\ &= \frac{(12)(20)}{42} \\ &= 5.714 \, \Omega \end{aligned}$$

$$\begin{aligned} R_2 &= \frac{(10)(20)}{12 + 20 + 10} \\ &= \frac{(10)(20)}{42} \\ &= 4.762 \, \Omega \end{aligned}$$

$$\begin{aligned} R_3 &= \frac{(12)(10)}{12 + 20 + 10} \\ &= \frac{(12)(10)}{42} \\ &= 2.857 \, \Omega \end{aligned}$$

Step 4 of 6

Determine the equivalent resistance up to $35 \, \Omega$ resistance.

$$\begin{aligned} R_{\text{eq1}} &= [(15 + R_3) \parallel (30 + R_2)] + R_1 \\ &= [(15 + 2.857) \parallel (30 + 4.762)] + 5.714 \\ &= (17.857 \parallel 34.762) + 5.714 \\ &= \frac{(17.857)(34.762)}{17.857 + 34.762} + 5.714 \\ &= 11.797 + 5.714 \\ &= 17.511 \, \Omega \end{aligned}$$

Redraw the circuit by replacing the resistor combination up to $35 \, \Omega$ by its equivalent resistance, $17.511 \, \Omega$.

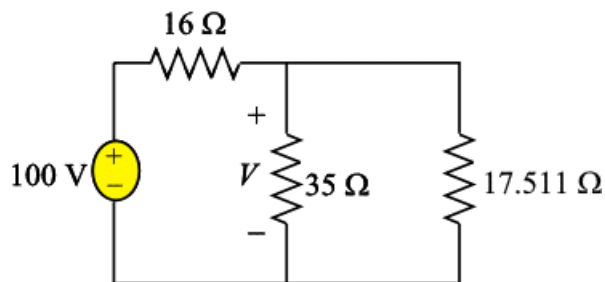


Figure 3

Step 5 of 6

Determine the equivalent resistance of the parallel combination of two resistances, $35\ \Omega$ and $17.511\ \Omega$.

$$\begin{aligned} R_{eq2} &= 35 \parallel 17.511 \\ &= \frac{(35)(17.511)}{35 + 17.511} \\ &= 11.672\ \Omega \end{aligned}$$

Redraw the circuit in Figure 3 by replacing the parallel resistances, $35\ \Omega$ and $17.511\ \Omega$ by their equivalent resistance.

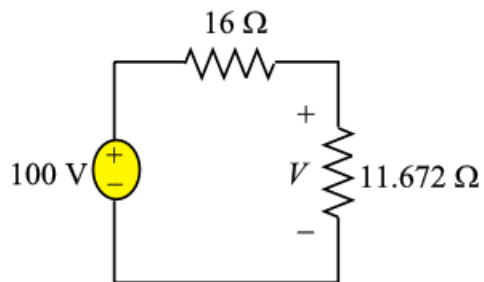


Figure 4

Step 6 of 6

Apply voltage division rule to find the voltage V .

$$\begin{aligned} V &= (100) \left(\frac{11.672}{16 + 11.672} \right) \\ &= (100) \left(\frac{11.672}{27.672} \right) \\ &= 42.18\ \text{V} \end{aligned}$$

Thus, the voltage V in the circuit is $42.18\ \text{V}$.